Midterm Report

Title: How well can school socioeconomic, geographic, and organizational characteristics predict AP participation in U.S. public high schools, and which characteristics are most predictive?

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Public Github Repo: https://github.com/molinazhang/learning_curve_ds1030.git

Introduction:

Advanced Placement (AP) courses provide high school students with opportunities to study college-level material and potentially earn college credit through qualifying exam scores. AP participation is associated with students' college pathways. Chajewski et al. (2011) found that AP exam participation predicted four-year college enrollment, while Mattern et al. (2009) found that students with qualifying AP scores attended more selective institutions, even after accounting for SAT scores and high school GPA. However, opportunities to participate in AP are unevenly distributed. The U.S. Government Accountability Office (2018) found that students attending smaller and less well-funded high schools had less access to advanced coursework, including AP. Together, these findings motivate examining how AP participation varies across schools and which school characteristics predict those differences. Such an analysis can help describe patterns of unequal participation in advanced coursework, shedding light on inequality in access to education in the United States.

Previous research has documented differences in AP access and participation across socioeconomic and demographic contexts. Xu, Solanki, and Fink (2021) combined Civil Rights Data Collection (CRDC), Common Core of Data (CCD), and other sources to examine racial participation gaps across school districts. They found large White-Black and White-Hispanic gaps in AP enrollment in most districts. Similarly, Barnard-Brak et.al. (2011) found that school demographic composition, including the shares of rational/ethnic minority and low-income students, was more strongly associated with whether schools offered AP courses than with how many students enrolled. However, prior studies have examined which factors are associated with AP participation or AP availability, but few have assessed how much the variation of AP participation can be explained, or predicted, by observed school characteristics.

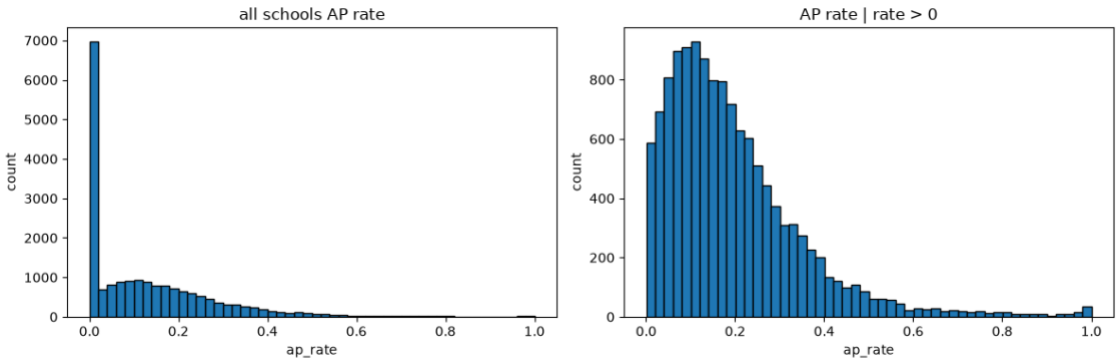
Our project will combine school characteristics from the Common Core of Data (CCD) with AP enrollment data from the Civil Rights Data Collection (CRDC) to address the following questions using supervised machine learning: How well can school socioeconomic, geographic, and organizational characteristics predict AP participation in U.S. public high schools, and which characteristics are most predictive? We will use a two-stage supervised learning approach. Stage 1 is binary classification aimed at predicting whether a school has any AP enrollment. Stage 2 is regression aimed at predicting the AP participation rate among schools with positive AP enrollment. We will compare logistic regression and OLS baselines with random forest and gradient boosting models. Hyperparameters will be tuned using cross-validation within the training set, reserving a held-out test set for final evaluation. AP enrollment counts will be excluded from predictors to prevent target leakage.

Data:

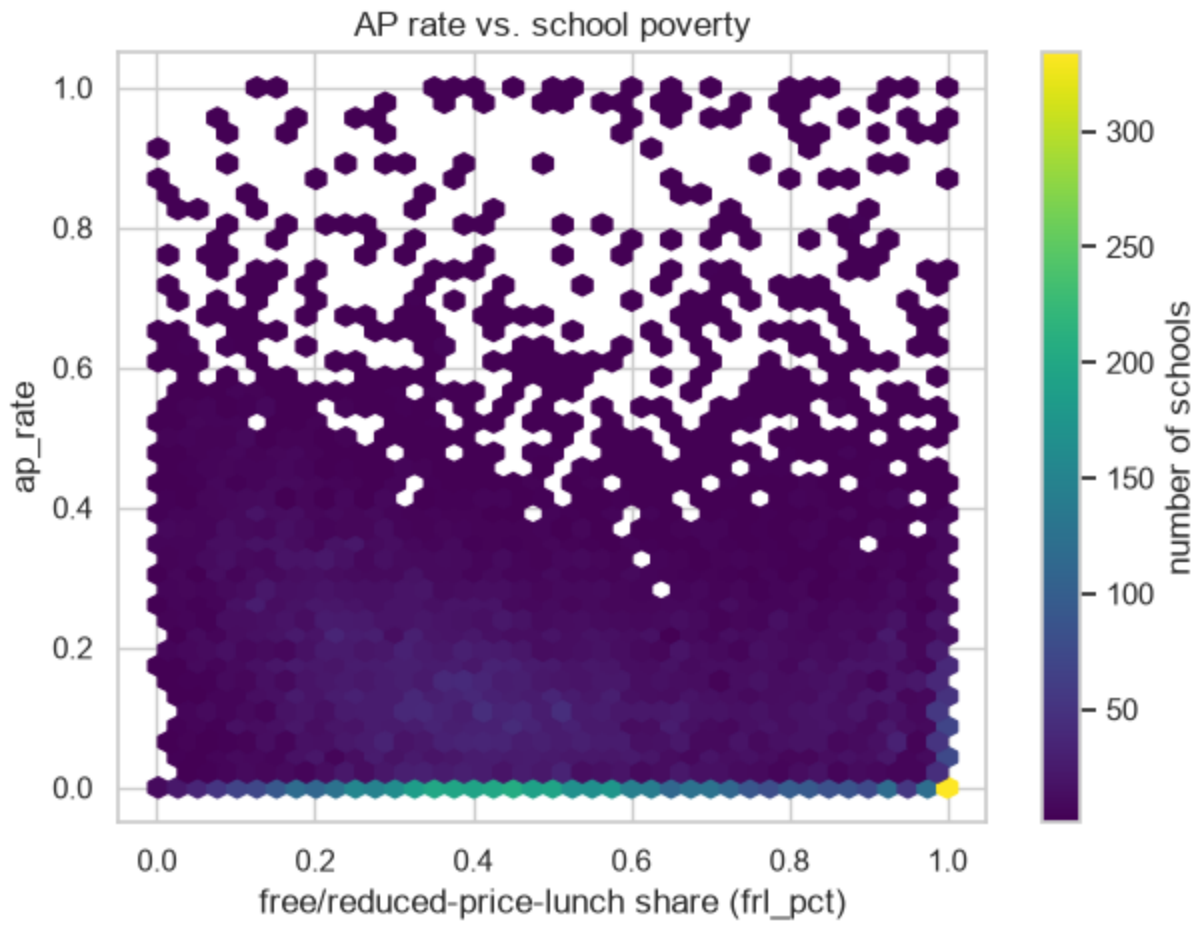
We collected 2021 school data through the Urban Institute's Education Data Portal API. We combined school directory and grades 9–12 enrollment data from the National Center for Education Statistics' Common Core of Data (CCD), and AP enrollment totals from the Department of Education's Civil Rights Data Collection (CRDC). The cleaned dataset contains 19,416 rows and 20 columns, with one row per regular public school. It covers the 50 states of the U.S., Washington, DC, and Puerto Rico, excluding virtual schools. Variables include numerical counts, coordinates, and ratios, nominal categories such as locale and school type, and the ordinal lowest grade offered. Important predictors include free or reduced-price lunch eligibility as a socioeconomic proxy, school enrollment, student–teacher ratio, location, and charter, magnet, and Title I status. School identifiers are excluded from predictors. We define two target variables - whether AP enrollment is positive for classification, and AP enrollment divided by grades 9–12 enrollment for regression among schools with an AP enrollment.

Exploratory Data Analysis:

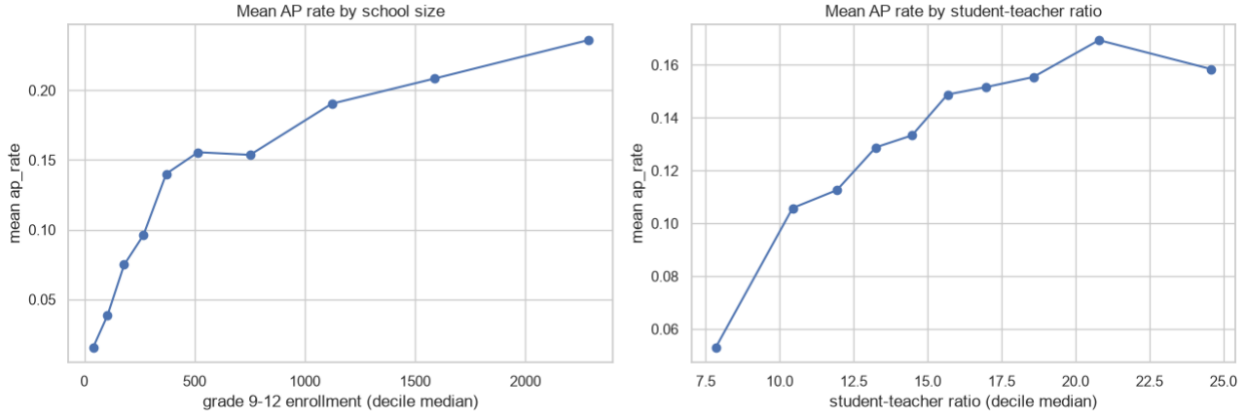
We conducted exploratory data analysis to understand the distribution of AP participation and how it may relate to school characteristics. We first used histograms to examine the distribution of AP participation, the target feature, which is the number of AP-enrolled students divided by the number of students in grades 9-12. We found its distribution is bounded between 0 and 1, with approximately 33 % of schools having an AP participation rate of zero. Since the target feature is zero-flated, we showed the distribution of AP rates both for all schools and for schools with an AP rate greater than zero.



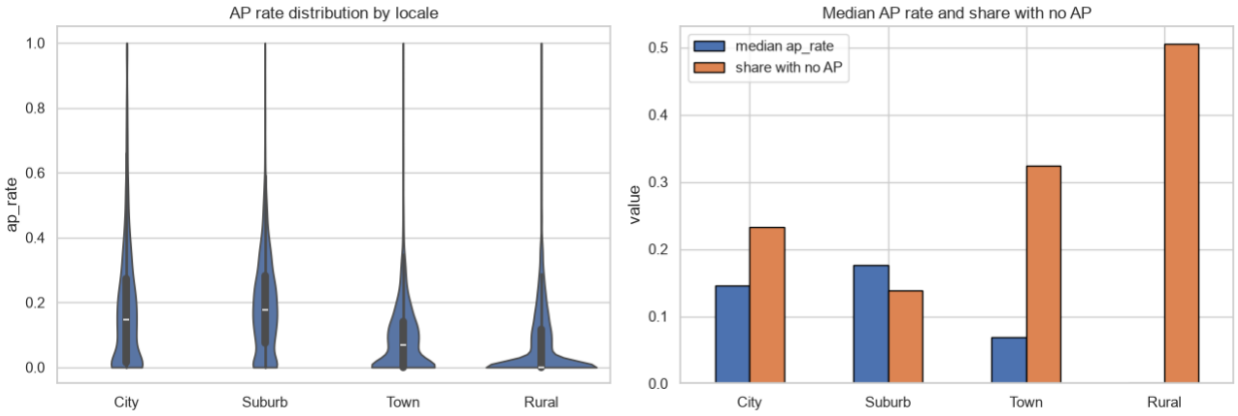
We then explored how AP rates are related to school characteristics. First, we used a hexbin plot to examine the relationship between the free or reduced price lunch share and AP participation rate, considering these numerical characteristics. Because the dataset contains many schools, plotting every school as a separate point could cause substantial overlap. The hexbin plot groups schools into hexagonal bins and uses color to show the number of schools in each bin, making the density patterns easier to see.



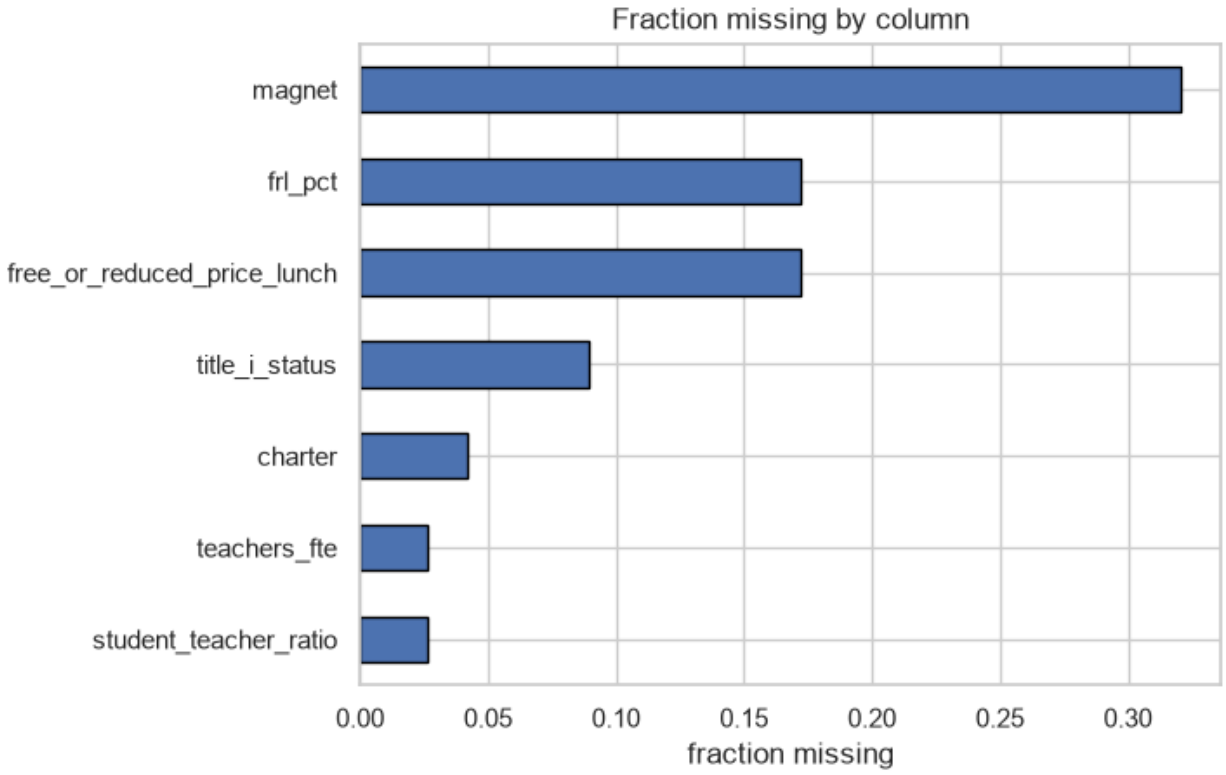
We also divided school size and student-teacher ratio into deciles and plotted the mean AP rate for each bin to show overall trends.



For categorical features, we used violin plots to compare the AP-rate distributions at a school-level across locale types (city, suburb, town, and rural locales). In the same figure, a bar chart compares each locale's median AP rate with the share of schools whose AP rate is zero. These two measures are well suited to distinguish a locale's typical level of AP participation from the prevalence of schools with no AP participation.



Finally, we checked missing-value rates and correlations among numerical features to identify data-quality issues and relationships that could affect modeling. For example, the lunch-share and AP-rate variables had a weak negative correlation, while the binned pilot showed that larger schools generally had higher mean AP participation.



Through exploratory data analysis, we examined how school characteristics, such as enrollment size and student demographics, relate to AP participation through exploratory data analysis.

Problem description and potential impact:

Beyond evaluating predictive accuracy, this project aims to identify patterns in school and region features associated with low AP participation. District administrators and educational

organizations could use this model to estimate AP participation rates from available school characteristics and flag schools for further review. Staff could investigate potential barriers for low AP participation rates, and use that information to guide discussions about teacher training and course expansions. By automating an initial screening process across thousands of schools, the pipeline could help organizations prioritize their analytical and outreach efforts more efficiently.

References:

Barnard-Brak, L., McGaha-Garnett, V., & Burley, H. (2011). Advanced Placement Course Enrollment and School-Level Characteristics. *NASSP Bulletin*, *95*(3), 165-174. https://doi-org.brown.idm.oclc.org/10.1177/0192636511418640

Chajewski, M., Mattern, K. D., & Shaw, E. J. (2011). Examining the role of Advanced Placement exam participation in 4-year college enrollment. *Educational Measurement: Issues and Practice*, *30*(4), 16-27.

Mattern, K. (2009). The relationship between AP exam performance and college outcomes.

Xu, D., Solanki, S., & Fink, J. (2021). College acceleration for all? Mapping racial gaps in advanced placement and dual enrollment participation. *American Educational Research Journal, 58*(5), 954–992. https://doi.org/10.3102/0002831221991138

U.S. Government Accountability Office. (2018). *K-12 education: Public high schools with more students in poverty and smaller schools provide fewer academic offerings to prepare for college* (GAO-19-8). https://www.gao.gov/products/gao-19-8